

“AI-Powered Humanoid Robot Fleet Management System”

Team: Robo-Nexus

Team Members:

Kiran Khatri	kkhatri@sfsu.edu
Kerry Yu	kyu19@sfsu.edu
Dias Almat	dalmat@sfsu.edu
Ishaank Zalpuri	izalpuri@sfsu.edu
Etienne Ghashehbaba	eghashehbaba@mail.sfsu.edu

GitHub Repository:

<https://github.com/sfsu-joseo/csc675-775-database-api-project-robo-nexus>

Version	Description
V1	initial submission for review

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Project Description

This project focuses on designing a database system to manage fleets of humanoid robots used in environments such as warehouses, hospitals, and homes. As humanoid robots become more common, they generate a large amount of information, including task assignments, task execution history, deployment locations, energy levels, maintenance records, and interaction details. Currently, much of this information is either spread across different tools or not stored in a structured way, making it difficult to track robot behavior and performance over time.

There are existing robot fleet management platforms, such as `Formant` and `InOrbit`, that help monitor robots and show their current status. These systems are useful for real-time monitoring and coordination, but they are mainly designed for general robot fleets and do not focus on organizing long-term data for humanoid robots. In particular, they do not emphasize storing detailed interaction context, coordination roles, or historical data in a centralized database.

One unique feature of the system is its ability to store *human emotional context data* in personal and healthcare environments. In settings such as hospitals or homes, humanoid robots often interact closely with people. The system allows simple emotional observations, such as calm or urgent situations, to be recorded as data for later analysis. Most existing fleet management tools do not capture or organize this type of information.

Another unique feature is *role-based robot coordination* in warehouse environments. The system allows certain robots to be assigned a *senior* role to help coordinate tasks among other robots based on factors such as hardware capability and energy level. If a senior robot becomes unavailable, the system supports reassigning that role to another robot, helping prevent disruptions. Existing tools typically treat all robots equally and do not support this type of structured coordination with a fail-safe mechanism.

Overall, this project provides a centralized and structured way to manage humanoid robot data that existing solutions do not fully address. Manufacturers, warehouses, hospitals, and households deploying humanoid robots can benefit from a system that keeps robot information organized, reliable, and easy to understand over time.

Functional Database Requirements

1. Robot

- 1.1. A robot shall belong to exactly one robot model.
- 1.2. A robot shall be deployed at at most one deployment location at a time.
- 1.3. A robot shall execute many tasks over time.
- 1.4. A robot shall be assigned at most one role at a time.
- 1.5. A robot shall have many emotion records over time.
- 1.6. A robot shall have at most one operator.
- 1.7. A robot shall have many energy status records over time.
- 1.8. A robot shall have many sensors
- 1.9. A robot shall have many charging stations

2. Robot Model

- 2.1. A robot model shall be produced by exactly one manufacturer.
- 2.2. A robot model shall be associated with many robots.
- 2.3. A robot model shall store the hardware and capability constraints of robots.

3. Manufacturer

- 3.1. A manufacturer shall produce one or more robot models.
- 3.2. A manufacturer shall have a unique identifier.

4. Deployment Location

- 4.1. A deployment location shall host many robots
- 4.2. A deployment location shall support different environment types (warehouse, hospital, household).
- 4.3. A deployment location shall host many operators
- 4.4. A deployment location shall have many logs
- 4.5. A deployment location shall have many energy stats

5. Task

- 5.1. A task shall be defined independently of any robot.
- 5.2. A task shall be assigned to many robots.

5.3. A task shall have a priority level.

6. Task Assignment

- 6.1. A task assignment shall associate exactly one robot with one task.
- 6.2. A task assignment shall record the time at which a task was assigned.
- 6.3. A task assignment shall support reassignment of tasks over time.

7. Task Execution

- 7.1. A task execution shall record the execution of one task by one robot.
- 7.2. A task execution shall have a start time and end time.

8. Operator (User)

- 8.1. An operator shall interact with many robots.
- 8.2. An operator shall have one or more roles.
- 8.3. An operator shall be assigned many deployment locations.

9. Emotion Record

- 9.1. An EmotionRecord shall store the observed emotional state associated with a human–robot interaction.
- 9.2. An EmotionRecord shall be associated with exactly one robot.
- 9.3. An EmotionRecord shall record the time at which the observation was made.
- 9.4. An EmotionRecord shall be associated with one deployment location where the interaction occurred.

10. Robot Role

- 10.1. A robot role shall be assigned to one or more robots.
- 10.2. A robot role shall represent a coordination role such as senior or worker.
- 10.3. A robot role may be reassigned to different robots over time.

11. Energy Status

- 11.1. An energy status shall be associated with exactly one robot.
- 11.2. An energy status shall record the battery level of the robot.
- 11.3. An energy status shall be associated with at most one deployment location

12. Log

- 12.1. A robot shall generate many logs.
- 12.2. A log shall be associated with one robot.
- 12.3. A log shall record operational events.
- 12.4. A log shall be generated by one sensor
- 12.5. A log shall be associated with at most one deployment location

13. Sensor

- 13.1. A sensor shall be associated with one robot
- 13.2. A sensor shall generate at least one log

14. Charging Station

- 14.1. A charging station shall serve many robots
- 14.2. A charging station shall record many charging sessions

15. Charging Session

- 15.1. A charging session shall occur at one charging station
- 15.2. A charging session shall be associated with one robot

16. Support Request

17. AI

Non-Functional Database Requirements

1. The system shall support concurrent access by at least n robots without performance degradation.
2. The system shall retain operational logs for at least 12 months.
3. The system shall be able to handle concurrent usage by _ robots
4. Any logs and data shall only be able to be accessed by an user with sufficient permissions
5. The system shall be able to maintain performance while receiving AI driven queries for real time data
6. (MAYBE)The system shall host AI queries and structural queries separately
- 7.